

Lab Assignment 3

[Control Flow Structure]

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Question1: For any character entered through keyboard check whether the entered character is capital letter, small letter, a digit or special symbol.

```
#include <stdio.h>
```

```
int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);

    if(ch >= 'A' && ch <= 'Z')
    {
        printf("Capital letter\n");
    }
    else if(ch >= 'a' && ch <= 'z')
    {
        printf("Small letter\n");
    }
    else if(ch >= '0' && ch <= '9')
    {
        printf("Digit\n");
    }
    else
    {
        printf("Special symbol");
    }

    return 0;
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc charcheck.c -o charcheck
atharva@quasar-2 Lab Assignment 3 % ./charcheck
Enter a character: 3
Digit%
atharva@quasar-2 Lab Assignment 3 % ./charcheck
Enter a character: A
Capital letter%
```

Question2: To find the greatest of 3 integers using two methods (a) if-else and (b) ternary operator.

2-a.

```
#include<stdio.h>
int main(){
    int a, b, c;

    printf("\nEnter the 3 numbers: ");
    scanf("%d", &a);
    scanf("%d", &b);
    scanf("%d", &c);

    if(a > b && a > c){
        printf("\n%d is the greatest integer.", a);
    }
    else if(b > a && b > c){
        printf("\n%d is the greatest integer.", b);
    }
    else{
        printf("\n%d is the greatest integer.", c);
    }

    return 0;
}
```

```
atharva@quasar-2 Lab Assignment 3 % ./bigcheck
```

```
Enter the 3 numbers: 2
```

```
3
```

```
4
```

```
4 is the greatest integer.%
```

2-b:

//method 2

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, c, largest;
```

```
    printf("Enter three integers: ");
```

```
    scanf("%d %d %d", &a, &b, &c);
```

```
    largest = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
```

```
printf("Largest int = %d\n", largest);
```

```
return 0;
```

```
}
```

```
atharva@quasar-2 Lab Assignment 3 % ./bigcheck2
```

```
Enter three integers: 2 5 10
```

```
Largest int = 10
```

```
atharva@quasar-2 Lab Assignment 3 % 
```

Question3: To implement calculator (+, -, *, %) using switch statements.

```
#include<stdio.h>
int main(){
    float num1, num2;
    float res;
    char op;

    printf("\nEnter two numbers: ");
    scanf("%f", &num1);
    scanf("%f", &num2);

    printf("\nEnter the operator: ");
    scanf(" %c", &op);

    switch(op){
        case '+':
            res = num1 + num2;
            printf("\n%f + %f = %f", num1, num2, res);
            break;

        case '-':
            res = num1 - num2;
            printf("\n%f - %f = %f", num1, num2, res);
            break;

        case '*':
            res = num1 * num2;
            printf("\n%f x %f = %f", num1, num2, res);
            break;

        case '/':
            res = num1 / num2;
            printf("\n%f / %f = %f", num1, num2, res);
            break;

        default:
            printf("\nPlease enter valid input.\n");
            break;
    }
    return 0;
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc cac1.c -o calc
atharva@quasar-2 Lab Assignment 3 % ./calc

Enter two numbers: 20 30

Enter the operator: +

20.000000 + 30.000000 = 50.000000%
atharva@quasar-2 Lab Assignment 3 %
```

Question4: Display month name according to the month number using Switch Statement.

```
#include<stdio.h>
int main(){
    int n;

    printf("\nWhich month?");
    scanf("%d", &n);

    switch(n){
        case 1:
            printf("\nJanuary");
            break;
        case 2:
            printf("\nFebruary");
            break;
        case 3:
            printf("\nMarch");
            break;
        case 4:
            printf("\nApril");
            break;
        case 5:
            printf("\nMay");
            break;
        case 6:
            printf("\nJune");
            break;
        case 7:
            printf("\nJuly");
            break;
        case 8:
            printf("\nAugust");
            break;
        case 9:
            printf("\nSeptember");
            break;
        case 10:
            printf("\nOctober");
            break;
        case 11:
            printf("\nNovember");
            break;
        case 12:
            printf("\nDecember");
            break;
    }
```

```
        default:
            printf("Invalid Input");
            break;
    }
    return 0;
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc month.c -o month
atharva@quasar-2 Lab Assignment 3 % ./month
```

```
Which month?12
```

```
December%
```

```
atharva@quasar-2 Lab Assignment 3 %
```

Question5: To check whether a number is: (a) even or odd (b) negative or positive by using if else.

```
#include <stdio.h>
```

```
int main(){
```

```
    int num;
```

```
    printf("Enter a number: \n");
```

```
    scanf("%d", &num);
```

```
    if(num < 0){
```

```
        printf("%d is negative. \n", num);
```

```
    }
```

```
    else if(num > 0){
```

```
        printf("%d is positive. \n", num);
```

```
    }
```

```
    else{
```

```
        printf("%d is 0. \n", num);
```

```
    }
```

```
    if(num % 2 == 0){
```

```
        printf("%d is even \n", num);
```

```
    }
```

```
    else if(num % 2 != 0){
```

```
        printf("%d is odd \n", num);
```

```
    }
```

```
    else{
```

```
        printf("invalid input");
```

```
    }
```

```
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc numcheck.c -o numcheck
```

```
atharva@quasar-2 Lab Assignment 3 % ./numcheck
```

```
Enter a number:
```

```
2
```

```
2 is positive.
```

```
2 is even
```

```
atharva@quasar-2 Lab Assignment 3 %
```

Question6: To check a no. say 'n' divisible by 5 and/or 8. Print the appropriate message accordingly.

```
#include<stdio.h>
int main(){
    int num;

    printf("enter number: \n");
    scanf("%d", &num);

    if(num % 5 == 0 && num % 8 == 0)
        printf("\n%dis divisible by both 5 and 8. ", num);

    else if(num % 5 == 0 && num % 8 != 0)
        printf("\n%d is divisible by 5 but not 8.", num);

    else if(num % 5 != 0 && num % 8 == 0)
        printf("\n%d is divisible by 8 but not 5.", num);

    else{
        printf("\n%d is neither divisible by 5 nor by 8.", num);
    }
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc div.c -o div
atharva@quasar-2 Lab Assignment 3 % ./div
enter number:
40

40is divisible by both 5 and 8.%
atharva@quasar-2 Lab Assignment 3 % ./div
enter number:
43

43 is neither divisible by 5 nor by 8.%
atharva@quasar-2 Lab Assignment 3 %
```

Question7: A library charges a fine for every book returned late. For the first 5 days the fine is 50 paisa, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be canceled. WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

```
#include<stdio.h>
int main(){
    int days;
    float fine;

    printf("\nHow many days have you delayed?");
    scanf("%d", &days);

    if(days < 5){
        printf("\nYour total fine is: ");
        fine = 0.5 * days;
        printf(" %f", fine);
    }
    else if(days > 5 && days < 10){
        printf("\nYour total fine is: ");
        fine = 1 * days;
        printf(" %f", fine);
    }
    else if(days > 10){
        printf("\nYour total fine is: ");
        fine = 5 * days;
        printf(" %f", fine);
    }
    else{
        printf("Invalid input");
    }
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc library.c -o library
atharva@quasar-2 Lab Assignment 3 % ./library

How many days have you delayed?3

Your total fine is:  1.500000%
atharva@quasar-2 Lab Assignment 3 % ./library

How many days have you delayed?23

Your total fine is:  115.000000%
atharva@quasar-2 Lab Assignment 3 %
```

Question8:

Write a program to read in 5 numbers and compute the average, max, min, standard deviation.

```
#include<stdio.h>
#include<math.h>
int main(){
    float a, b, c, d, e;

    printf("\nEnter the 5 values spearated by space.");
    scanf("%f", &a);
    scanf("%f", &b);
    scanf("%f", &c);
    scanf("%f", &d);
    scanf("%f", &e);

    float mean, min, max, std_dev;

    mean = (a + b + c + d +e)/5;

    printf("\nMean = %f", mean);

    min = a;
    max = a;

    if(b < min)
        min = b;

    if(c < min)
        min = c;

    if(d < min)
        min = d;

    if(e < min)
        min = e;

    if(b > max)
        max = b;

    if(c > max)
        max = c;

    if(d > max)
        max = d;
```

```

    if(e > max)
        max = e;

    printf("Minimum = %f\n", min);
    printf("Maximum = %f\n", max);

    std_dev = sqrt(
        ((a - mean) * (a - mean) +
        (b - mean) * (b - mean) +
        (c - mean) * (c - mean) +
        (d - mean) * (d - mean) +
        (e - mean) * (e - mean)) / 5
    );

    printf("standard deviation = %f", std_dev);
}
atharva@quasar-2 Lab Assignment 3 % gcc stddev.c -o stddev
atharva@quasar-2 Lab Assignment 3 % ./stddev

Enter the 5 values speparated by space.45 36 12 08 9

Mean = 22.000000Minimum = 8.000000
Maximum = 45.000000
standard deviation = 15.427249%
atharva@quasar-2 Lab Assignment 3 % █

```

Question9:

9. Using if-else decision making, write a program to read number of computers that need services from the user and calculate the service charge as per the information shown below.

PCs	Charges / per pc
Up to 3	100
4-6	75
7-9	50
>10	40

```
#include<stdio.h>
int main(){
    int num;

    printf("\nHow many computers need servicing?");
    scanf("%d", &num);

    if( num <= 3)
        printf("\nTotal cost = ₹%d", num*100);
    else if( num > 3 && num < 7)
        printf("\nTotal cost = ₹%d", num*75 );
    else if( num > 7 && num < 10)
        printf("\nTotal cost = ₹%d", num*50);
    else if( num > 10)
        printf("\nTotal cost = ₹%d", num*40);
    else
        printf("\nInvalid Input");

}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc comprepair.c -o comprepair
atharva@quasar-2 Lab Assignment 3 % ./comprepair

How many computers need servicing?34

Total cost = ₹1360%
atharva@quasar-2 Lab Assignment 3 %
```

Question10: Using if-else decision making, write a C program that reads an integer number (1–12) representing a month of the year and displays the total number of days in that month. If the user enters a number outside the range of 1 to 12, display an appropriate error message.

11. Using if-else decision making, write a program in C that displays the recommended actions depending on the color of a traffic light.

Color	Recommended Actions
Red ('R' or 'r')	STOP
Yellow ('Y' or 'y')	CAUTION
Green ('G' or 'g')	GO
Default	Wrong Color Selected

```
#include <stdio.h>
```

```
int main() {  
    char color;  
  
    printf("Enter a color: ");  
    scanf("%c", &color);  
  
    if (color == 'R' || color == 'r'){  
        printf("\nSTOP");  
    } else if (color == 'Y' || color == 'y'){  
        printf("\nCAUTION");  
    } else if (color == 'G' || color == 'g'){  
        printf("\nGO");  
    } else {  
        printf("\nINVALID COLOR");  
    }  
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc light.c -o light  
atharva@quasar-2 Lab Assignment 3 % ./light  
Enter a color: g  
  
GO%  
atharva@quasar-2 Lab Assignment 3 %
```

12. A bank provides loans at a base interest rate of 15%. The bank also offers rebates under the following conditions:

- 2% rebate if the applicant is from a rural area.
- 2% rebate if the applicant belongs to the below-poverty-line (BPL) category.
- 2% rebate if the loan is taken for agriculture, fishery, or dairy purposes.

Write a C program that reads the details of a person (whether they are from a rural area, belong to BPL, and the purpose of the loan) and computes the final interest rate applicable to the loan.

```
#include<stdio.h>
int main(){
    float amt, fin_amt;
    char ans1, ans2, ans3;

    printf("\n How much do you want to borrow? ");
    scanf("%f", &amt);

    printf("\n Your total payable amount with interest of 15%% will be: ");
    fin_amt = amt + 0.15*amt;
    printf("\n Final amt without rebates: ₹%f", fin_amt);

    printf("\n Answer the following in y or n to get rebates!");

    printf("\n Are you from a rural area?");
    scanf(" %c", &ans1);

    if (ans1 == 'y' || ans1 == 'Y'){
        fin_amt = fin_amt - (0.02*amt);
    }
    else{
        printf("\n You're not eligible for this rebate!");
    }

    printf("\n Do you categorise Below Poverty Line?");
    scanf(" %c", &ans2);

    if (ans2 == 'y' || ans2 == 'Y'){
        fin_amt = fin_amt - (0.02*amt);
    }
    else{
        printf("\n You're not eligible for this rebate!");
    }

    printf("\n Are you taking loan for Agriculture, Fisheries, or Dairy?");
    scanf(" %c", &ans3);
    if (ans3 == 'y' || ans3 == 'Y'){
        fin_amt = fin_amt - (0.02*amt);
    }
}
```

```
else{
    printf("\n You're not eligible for this rebate!");
}

printf("\n Your total payable amount is: ₹%f", fin_amt);
```

```
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc loan.c -o loan
atharva@quasar-2 Lab Assignment 3 % ./loan
```

How much do you want to borrow? 1000

Your total payable amount with interest of 15% will be:

Final amt without rebates: ₹1150.000000

Answer the following in y or n to get rebates!

Are you from a rural area?y

Do you categorise Below Poverty Line?n

You're not eligible for this rebate!

Are you taking loan for Agriculture, Fisheries, or Dairy?y

Your total payable amount is: ₹1110.000000%

```
atharva@quasar-2 Lab Assignment 3 %
```

13. Write a C program using a switch statement to print the lowercase English word corresponding to a number entered by the user. If the input number n is between 1 and 9 (inclusive), print its English word in lowercase (e.g., "one" for 1, "two" for 2, etc.). For any number greater than 9, print "greater than 9".

```
#include<stdio.h>
int main(){
    int num;

    printf("\n Number between 1 to 9: ");
    scanf("%d", &num);

    switch(num){
        case 1:
            printf("\n ONE");
            break;

        case 2:
            printf("\n TWO");
            break;

        case 3:
            printf("\n THREE");
            break;

        case 4:
            printf("\n FOUR");
            break;

        case 5:
            printf("\n FIVE");
            break;

        case 6:
            printf("\n SIX");
            break;

        case 7:
            printf("\n SEVEN");
            break;

        case 8:
            printf("\n EIGHT");
            break;

        case 9:
            printf("\n NINE");
```

```
        break;

default:
    printf("\n INVALID INPUT");
    break;
}
}
```

```
atharva@quasar-2 Lab Assignment 3 % gcc numword.c -o numword
atharva@quasar-2 Lab Assignment 3 % ./numword
```

```
Number between 1 to 9: 4
```

```
FOUR%
```

```
atharva@quasar-2 Lab Assignment 3 % ./numword
```

```
Number between 1 to 9: 10
```

```
INVALID INPUT%
```

```
atharva@quasar-2 Lab Assignment 3 %
```